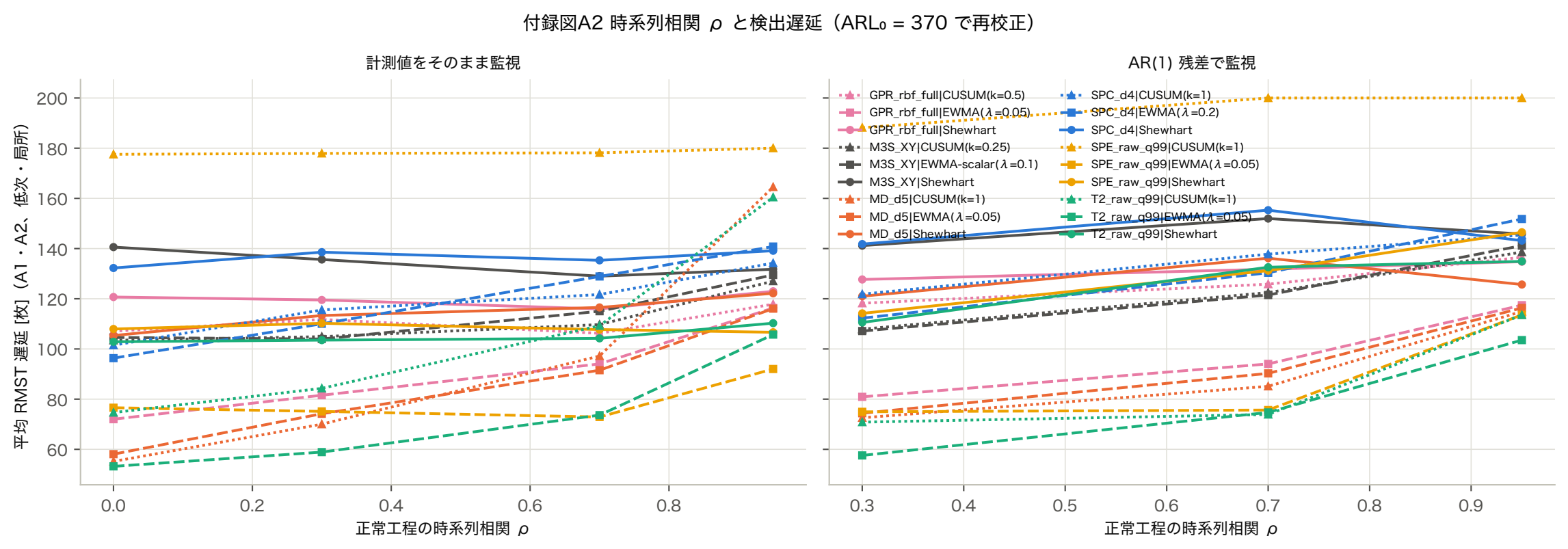
付録図A2 時系列相関 ρ と検出遅延（ARL₀ = 370 で再校正）
計測値をそのまま監視
AR(1) 残差で監視
平均 RMST 遅延 [枚]（A1・A2、低次・局所）
200
180
160
140
120
100
80
60
0.0
0.2
0.4
0.6
0.8
0.3
0.4
0.5
0.6
0.7
0.8
0.9
正常工程の時系列相関 ρ
正常工程の時系列相関 ρ
GPR_rbf_full|CUSUM(k=0.5)
GPR_rbf_full|EWMA(λ=0.05)
GPR_rbf_full|Shewhart
M3S_XY|CUSUM(k=0.25)
M3S_XY|EWMA-scalar(λ=0.1)
M3S_XY|Shewhart
MD_d5|CUSUM(k=1)
MD_d5|EWMA(λ=0.05)
MD_d5|Shewhart
SPC_d4|CUSUM(k=1)
SPC_d4|EWMA(λ=0.2)
SPC_d4|Shewhart
SPE_raw_q99|CUSUM(k=1)
SPE_raw_q99|EWMA(λ=0.05)
SPE_raw_q99|Shewhart
T2_raw_q99|CUSUM(k=1)
T2_raw_q99|EWMA(λ=0.05)
T2_raw_q99|Shewhart